

Chapter 18

RTC Timer

18.1 Introduction

RTC Timer is an important module for implementing low power management of ESP32-P4. Based on a 48-bit readable counter, RTC Timer is mainly used as a system timer in low power mode when the timer peripheral in the HP system is unavailable. It also allows for configuring timer interrupts and logging the time when specific events happen in the system. The supported events are described in the [18.3 Functional Description](#).

18.2 Feature List

RTC Timer has the following features:

- 48-bit counter
- Time logging when one of the following events happens:
 - HP system reset
 - CPU enters stall state
 - CPU exits stall state
 - Crystal powers up
 - Crystal powers down
- Time logging through register configuration
- Occurrence time cached of the most recent two specific events
- Generation of interrupts at target times, which are configurable. It is also possible to configure two target times simultaneously.
- Uninterrupted operation during any reset or sleep mode, except for power-on reset of LP system.

18.3 Functional Description

- 48-bit counter
 - The implementation of the RTC Timer is based on a 48-bit counter, driven by the RC_SLOW_CLK in the AlwaysOn power domain. It uses continuous loop counting (except during LP system reset), and an overflow interrupt is generated when the 48-bit counter overflows.
- Log the occurrence time of specific events
 - RTC Timer supports logging the occurrence time of three types of events:

- * HP system reset
 - * CPU enters or exits stall state
 - * Crystal powers up or down
- The above three types of events have their own independent configuration registers. When the configuration registers are enabled, if the corresponding events occur, the hardware pulse generated will trigger the RTC Timer to log the current time. The corresponding configuration registers are shown in the table below.

Table 18.3-1. Configure RTC Timer to Log the Occurrence Time of Specific Events

Configuration Register	Description
RTC_TIMER_MAIN_TIMER_SYS_RST	Enable the RTC Timer to log the time of system reset.
RTC_TIMER_MAIN_TIMER_SYS_STALL	Enable the RTC Timer to log the time when the CPU enters or exits the stall state.
RTC_TIMER_MAIN_TIMER_XTAL_OFF	Enable the RTC Timer to log the time when the crystal powers up or down.

- Log the current time through software configuration.
 - In addition to the above three specific events that can trigger the RTC Timer to log the time, the RTC Timer can also record the current time through the configuration register [RTC_TIMER_MAIN_TIMER_UPDATE](#).
 - A pulse signal will be generated by configuring the [RTC_TIMER_MAIN_TIMER_UPDATE](#), which will trigger the RTC Timer to record the current time.
- Generate counting interrupts at target times
 - RTC Timer supports configuring two target times simultaneously, referred to as target time 0 and target time 1. When the timer reaches target time 0, it will trigger the [RTC_TIMER_CMPO_INT](#) interrupt. Similarly, when the timer reaches Target Moment 1, it will trigger the [RTC_TIMER_CMP1_INT](#) interrupt.
 - Configure the target times through the registers below:

Table 18.3-2. Target times Configuration

Configuration Register	Description
RTC_TIMER_MAIN_TIMER_TAR_EN <i>n</i>	<i>n</i> = 0 or 1, enable target time configuration.
RTC_TIMER_MAIN_TIMER_TAR_LOW <i>n</i>	<i>n</i> = 0 or 1, configure the low 32 bits of the target time.
RTC_TIMER_MAIN_TIMER_TAR_HIGH <i>n</i>	<i>n</i> = 0 or 1, configure the high 16 bits of the target time.

- Read the time cached in the RTC Timer
 - There are two sets of registers used to cache the occurrence time of specific events, namely register group 0 and register group 1.
 - Register group 0 includes register [RTC_TIMER_MAIN_BUFO_LOW_REG](#) and [RTC_TIMER_MAIN_BUF1_HIGH_REG](#) which are used to cache the count value of the RTC Timer under the current trigger.

- Register group 1 includes register [RTC_TIMER_MAIN_BUF1_LOW_REG](#) and [RTC_TIMER_MAIN_BUF1_HIGH_REG](#), which are used to cache the count value of the RTC Timer from the last trigger.
- On a new trigger, the record from the previous trigger will be moved from register group 0 to register group 1 (and the previous record in register group 1 will be overwritten), and the record of this trigger will be stored in register group 0. Therefore, only the last two triggers can be logged at any time.

18.4 Event Task Matrix Feature

The RTC Timer on ESP32-P4 supports the Event Task Matrix (ETM) function, which allows RTC Timer's ETM tasks to be triggered by any peripherals' ETM events, or RTC Timer's ETM events to trigger any peripherals' ETM tasks. This section introduces the ETM tasks and events related to RTC Timer. For more information, please refer to Chapter [13 Event Task Matrix \(ETM\)](#).

RTC Timer doesn't receive any ETM tasks:

RTC Timer can generate the following ETM events:

- `RTC_EVT_CMP`: Indicate that the RTC Timer reaches the target time 0 configured by [RTC_TIMER_MAIN_TIMER_TAR_LOW0](#) and [RTC_TIMER_MAIN_TIMER_TAR_HIGH0](#).
- `RTC_EVT_OVF`: Indicate the 48-bit counter overflow.
- `RTC_EVT_TICK`: Indicates the event that occurs when the RTC timer increments by 1.

18.5 Interrupts

ESP32-P4's RTC Timer can generate the following interrupt signal(s) that will be sent to the [Interrupt Matrix](#).

- `LP_TIMER_REG_O_INTR`
- `LP_TIMER_REG_1_INTR`

There are several internal interrupt sources from RTC Timer that can generate the above interrupt signal(s). The interrupt sources from RTC Timer are listed with their trigger conditions and the resulted interrupt signal(s) in Table [18.5-1](#).

Table 18.5-1. RTC Timer's Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
<code>RTC_TIMER_CMPO_INT</code>	RTC Timer reaches the target time 0	<code>LP_TIMER_REG_O_INTR</code>
<code>RTC_TIMER_CMP1_INT</code>	RTC Timer reaches the target time 1	<code>LP_TIMER_REG_1_INTR</code>
<code>RTC_TIMER_OVERFLOW_INT</code>	48-bit counter overflow	<code>LP_TIMER_REG_O_INTR</code> <code>LP_TIMER_REG_1_INTR</code>

Note:

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter [12 Interrupt Matrix](#) > Section [12.2 Interrupt Terminology in ESP32-P4](#).

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section [18.6 Register Summary](#).

18.6 Register Summary

The addresses in this section are relative to RTC Timer base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

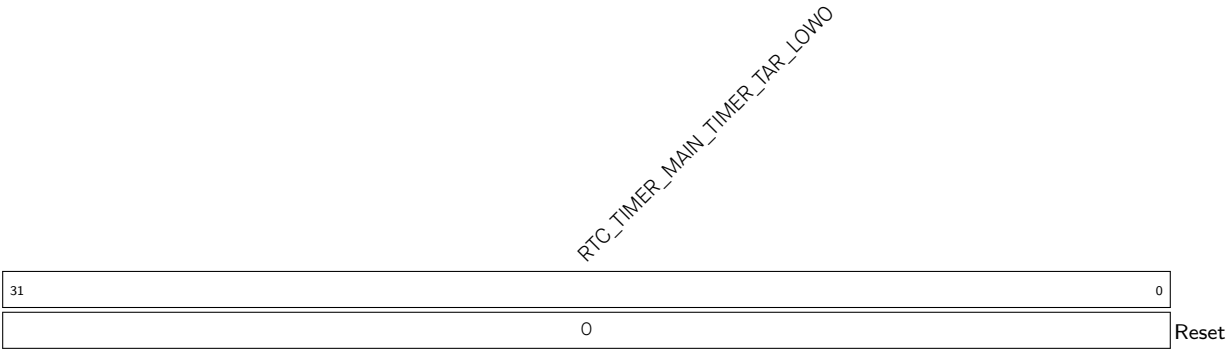
Name	Description	Address	Access
Configuration Registers			
RTC_TIMER_TARO_LOW_REG	Configure the low 32 bits of the target time 0	0x0000	R/W
RTC_TIMER_TARO_HIGH_REG	Configure the high 16 bits of the target time 0	0x0004	varies
RTC_TIMER_TAR1_LOW_REG	Configure the low 32 bits of the target time 1	0x0008	R/W
RTC_TIMER_TAR1_HIGH_REG	Configure the high 16 bits of the target time 1	0x000C	varies
RTC_TIMER_UPDATE_REG	Configure to enable the RTC Timer to latch current value	0x0010	varies
RTC_TIMER_MAIN_BUFO_LOW_REG	The low 32 bits of the cached value 0 in RTC Timer	0x0014	RO
RTC_TIMER_MAIN_BUFO_HIGH_REG	The high 16 bits of the cached value 0 in RTC Timer	0x0018	RO
RTC_TIMER_MAIN_BUF1_LOW_REG	The low 32 bits of the cached value 1 in RTC Timer	0x001C	RO
RTC_TIMER_MAIN_BUF1_HIGH_REG	The high 16 bits of the cached value 1 in RTC Timer	0x0020	RO
Interrupt Registers			
RTC_TIMER_INT_RAW_REG	The interrupt raw status register for the RTC Timer reaching the target time 0	0x0028	RO
RTC_TIMER_INT_ST_REG	The interrupt status register for the RTC Timer reaching the target time 0	0x002C	RO
RTC_TIMER_INT_ENA_REG	The interrupt enable register for the RTC Timer reaching the target time 0	0x0030	R/W
RTC_TIMER_INT_CLR_REG	The interrupt clear register for the RTC Timer reaching the target time 0	0x0034	WO
RTC_TIMER_LP_INT_RAW_REG	The interrupt raw status register for RTC Timer reaching the target time 1	0x0038	RO
RTC_TIMER_LP_INT_ST_REG	The interrupt status register for the RTC Timer reaching the target time 1	0x003C	RO
RTC_TIMER_LP_INT_ENA_REG	The interrupt enable register for the RTC Timer reaching the target time 1	0x0040	R/W
RTC_TIMER_LP_INT_CLR_REG	The interrupt clear register for the RTC Timer reaching the target time 1	0x0044	WO
Version Register			
RTC_TIMER_DATE_REG	Version control register	0x03FC	R/W

18.7 Registers

The addresses in this section are relative to RTC Timer base address provided in Table 7.3-2 in Chapter 7 [System and Memory](#).

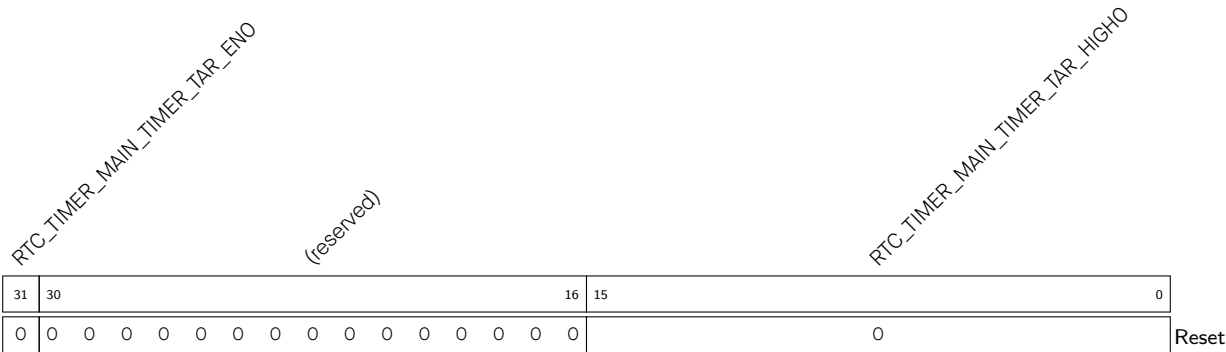
For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 18.1. RTC_TIMER_TARO_LOW_REG (0x0000)



RTC_TIMER_MAIN_TIMER_TAR_LOWO Configure the low 32 bits of the target time 0 of the RTC Timer. (R/W)

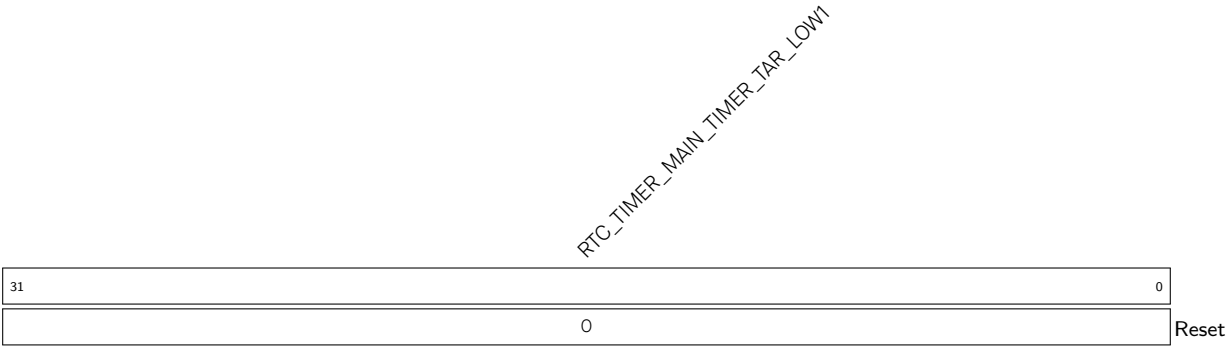
Register 18.2. RTC_TIMER_TARO_HIGH_REG (0x0004)



RTC_TIMER_MAIN_TIMER_TAR_HIGHO Configure the high 16 bits of the target time 0. (R/W)

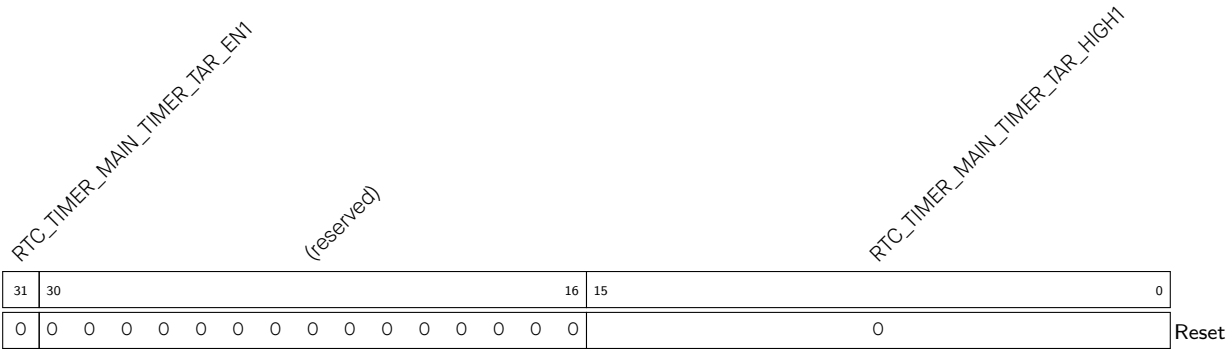
RTC_TIMER_MAIN_TIMER_TAR_ENO Configure to enable the target time 0 of the RTC Timer. (WO)

Register 18.3. RTC_TIMER_TAR1_LOW_REG (0x0008)



RTC_TIMER_MAIN_TIMER_TAR_LOW1 Configure the low 32 bits of the target time 1. (R/W)

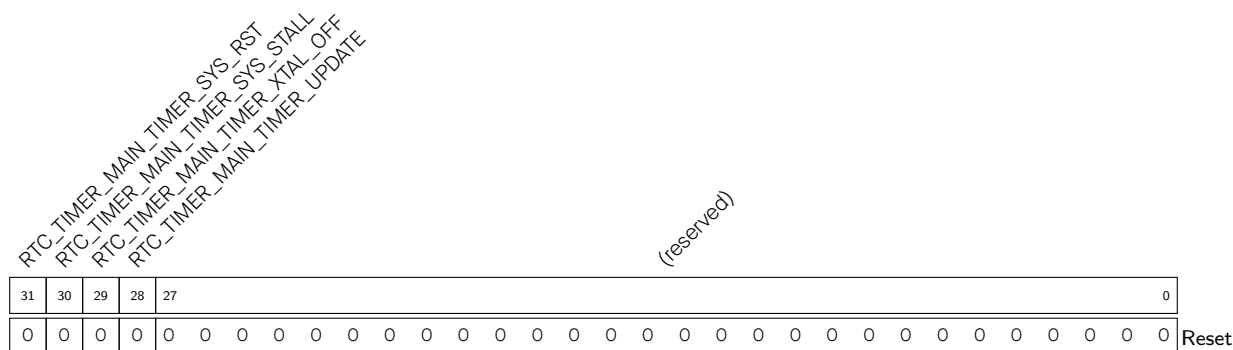
Register 18.4. RTC_TIMER_TAR1_HIGH_REG (0x000C)



RTC_TIMER_MAIN_TIMER_TAR_HIGH1 Configure the high 16 bits of the target time 1. (R/W)

RTC_TIMER_MAIN_TIMER_TAR_EN1 Configure to enable the target time 1 of the RTC Timer. (WO)

Register 18.5. RTC_TIMER_UPDATE_REG (0x0010)



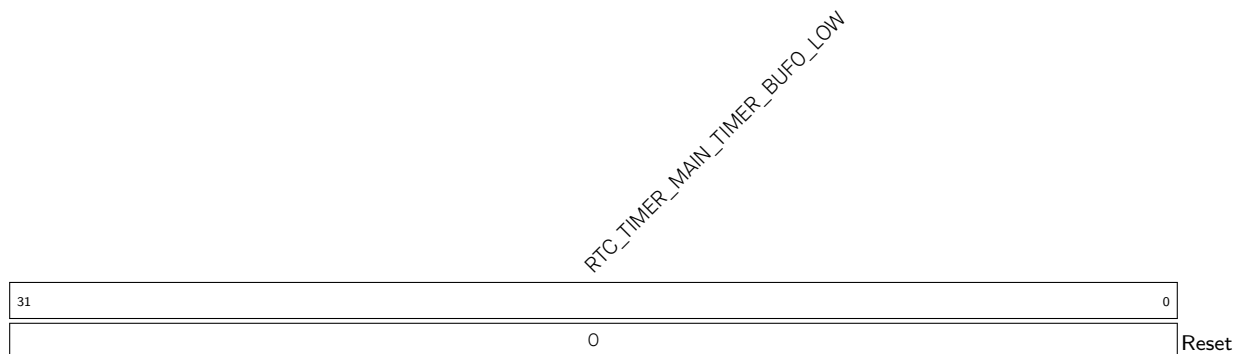
RTC_TIMER_MAIN_TIMER_UPDATE Configure to log the current time through software configuration. (WO)

RTC_TIMER_MAIN_TIMER_XTAL_OFF Configure to enable RTC Timer to log the time when the crystal powers up or down. (R/W)

RTC_TIMER_MAIN_TIMER_SYS_STALL Configure to enable the RTC Timer to log the time when the CPU enters or exits the stall state. (R/W)

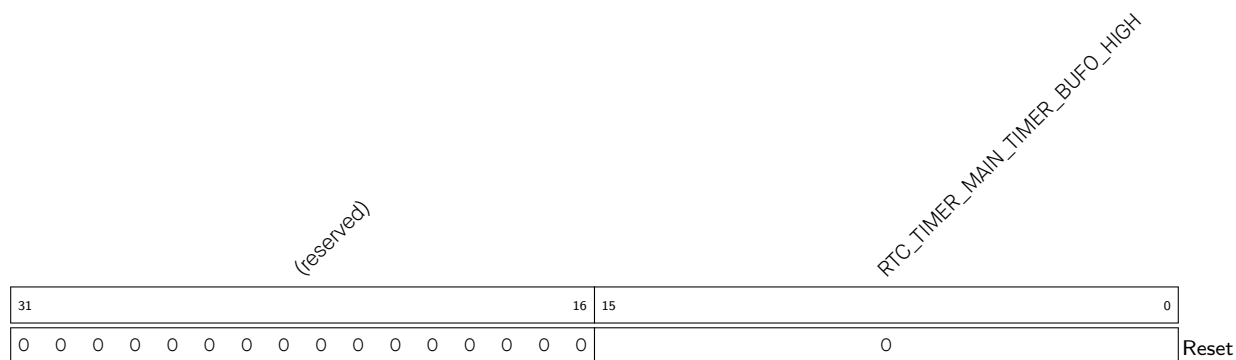
RTC_TIMER_MAIN_TIMER_SYS_RST	Configure to enable RTC Timer to log the time of system reset. (R/W)
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Register 18.6. RTC_TIMER_MAIN_BUFO_LOW_REG (0x0014)



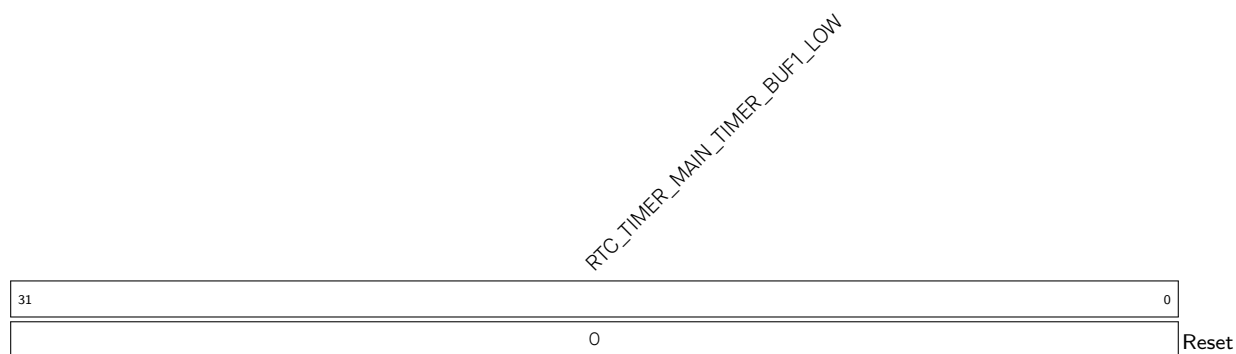
RTC_TIMER_MAIN_TIMER_BUFO_LOW Represent the low 32 bits of the cached value 0 in RTC Timer. (RO)

Register 18.7. RTC_TIMER_MAIN_BUFO_HIGH_REG (0x0018)



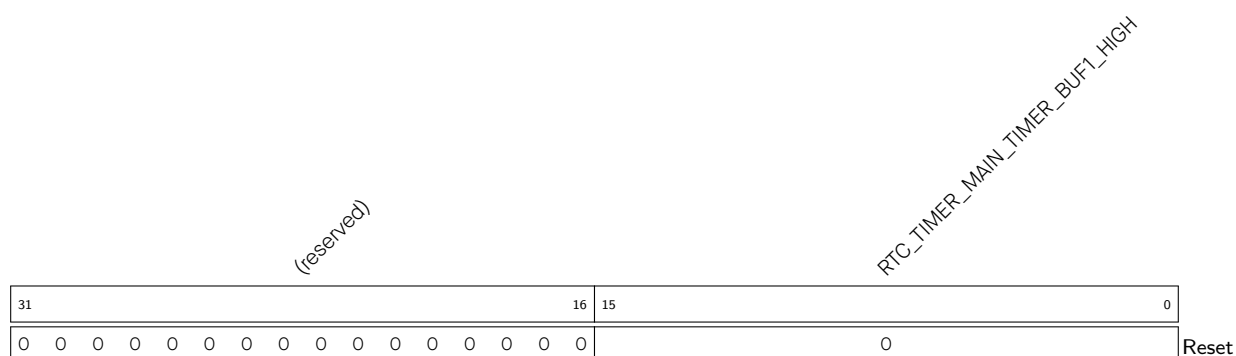
RTC_TIMER_MAIN_TIMER_BUFO_HIGH Represent the high 16 bits of the cached value 0 in RTC Timer. (RO)

Register 18.8. RTC_TIMER_MAIN_BUF1_LOW_REG (0x001C)



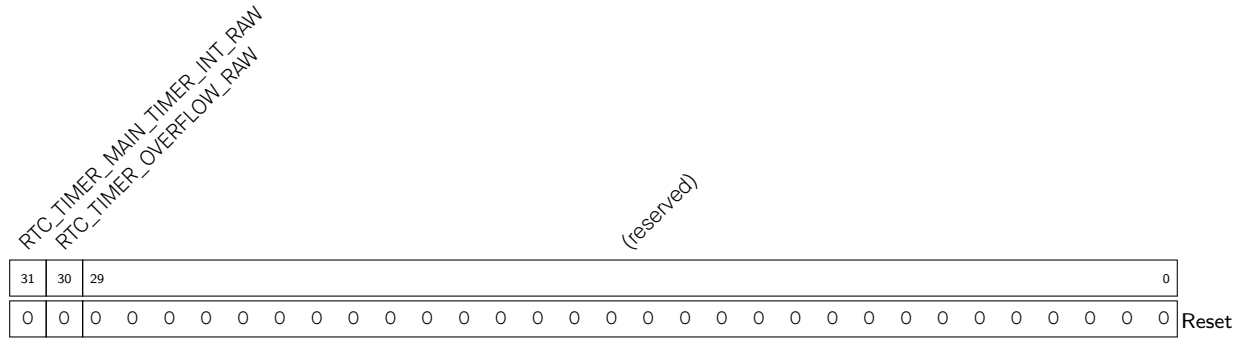
RTC_TIMER_MAIN_TIMER_BUF1_LOW Represent the low 32 bits of the cached value 1 in RTC Timer. (RO)

Register 18.9. RTC_TIMER_MAIN_BUF1_HIGH_REG (0x0020)



RTC_TIMER_MAIN_TIMER_BUF1_HIGH Represent the high 16 bits of the cached value 1 in RTC Timer. (RO)

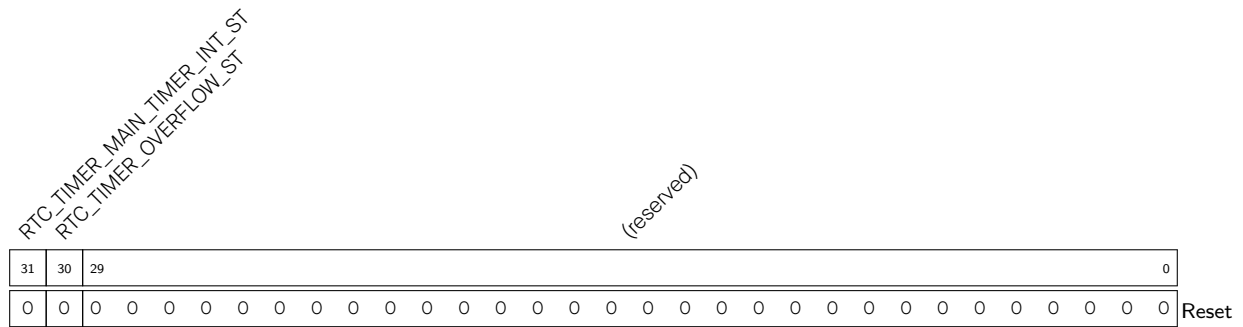
Register 18.10. RTC_TIMER_INT_RAW_REG (0x0028)



RTC_TIMER_OVERFLOW_RAW The interrupt raw status for the 48-bit counter overflow. (RO)

RTC_TIMER_MAIN_TIMER_INT_RAW The interrupt raw status for the RTC Timer reaching the target time 0. (RO)

Register 18.11. RTC_TIMER_INT_ST_REG (0x002C)



RTC_TIMER_OVERFLOW_ST The interrupt status for the 48-bit counter overflow. (RO)

RTC_TIMER_MAIN_TIMER_INT_ST The interrupt status for the RTC Timer reaching the target time 0. (RO)

Register 18.12. RTC_TIMER_INT_ENA_REG (0x0030)

Diagram illustrating the structure of the `RTC_TIMER_OVERFLOW_ENA` register. The register is 32 bits wide. Bit 31 is labeled `RTC_TIMER_MAIN_TIMER_INT_ENA`. Bit 30 is labeled `RTC_TIMER_OVERFLOW_ENA`. Bits 29 through 0 are labeled `(reserved)`.

RTC_TIMER_OVERFLOW_ENA Configure to enable the interrupt for 48-bit counter overflow. (R/W)

RTC_TIMER_MAIN_TIMER_INT_ENA Configure to enable the interrupt for the RTC Timer reaching the target time 0. (R/W)

Register 18.13. RTC_TIMER_INT_CLR_REG (0x0034)

RTC_TIMER_OVERFLOW_CLR

RTC_TIMER_MAIN_TIMER_INT_CLR

(reserved)

0

RTC_TIMER_OVERFLOW_CLR Configure to clear the interrupt for 48-bit counter overflow. (WO)

RTC_TIMER_MAIN_TIMER_INT_CLR Configure to clear the interrupt for the RTC Timer reaching the target time 0. (WO)

Register 18.14. RTC_TIMER_LP_INT_RAW_REG (0x0038)

Diagram of the RTC_TIMER_MAIN_TIMER_LP_INT_RAW register. The register is 32 bits wide. Bits 31 and 30 are labeled `RTC_TIMER_MAIN_TIMER_LP_INT_RAW`. Bit 29 is labeled `RTC_TIMER_MAIN_TIMER_OVERFLOW_LP_INT_RAW`. The remaining bits (0-28) are labeled `(reserved)`. The register is shown as a horizontal bar with a `Reset` button on the right.

RTC_TIMER_MAIN_TIMER_OVERFLOW_LP_INT_RAW The interrupt raw status for the 48-bit counter overflow. (RO)

RTC_TIMER_MAIN_TIMER_LP_INT_RAW The interrupt raw status for the RTC Timer reaching the target time 1.(RO)

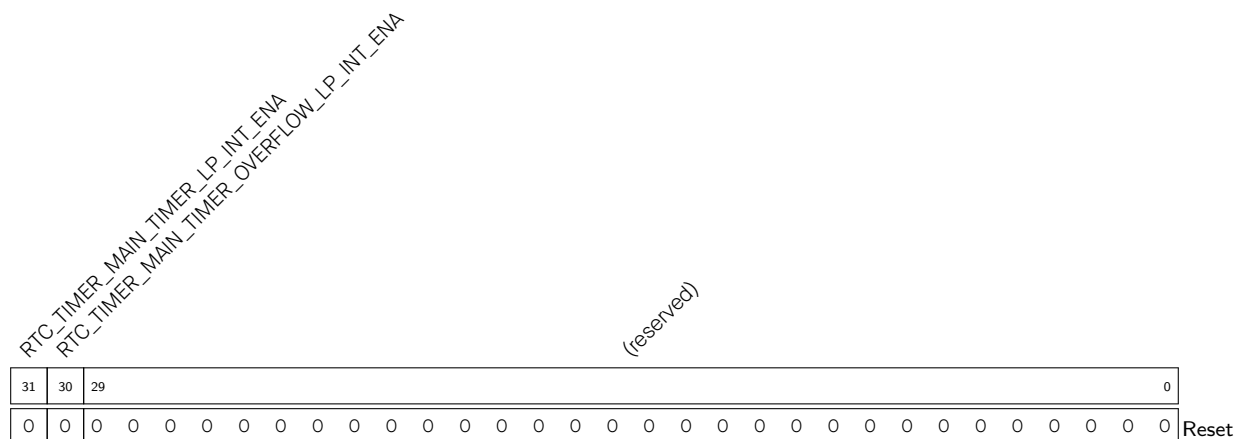
Register 18.15. RTC_TIMER_LP_INT_ST_REG (0x003C)

Diagram of the RTC_TIMER_MAIN_TIMER_LP_INT_ST register. The register is 32 bits wide. Bit 31 is labeled 'RTC_TIMER_MAIN_TIMER_LP_INT_ST'. Bit 30 is labeled 'RTC_TIMER_MAIN_TIMER_OVERFLOW_LP_INT_ST'. Bits 29-0 are labeled '(reserved)'. The register is shown as a horizontal bar with bit positions 31, 30, and 29 marked at the left end and 0 at the right end. Below the bar, the bits are shown as a sequence of 0s and 1s, with the last bit (bit 0) being 1 and the others being 0.

RTC_TIMER_MAIN_TIMER_OVERFLOW_LP_INT_ST	The interrupt status for the 48-bit counter overflow. (RO)
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RTC_TIMER_MAIN_TIMER_LP_INT_ST	The interrupt status for the RTC Timer reaching the target time 1.(RO)
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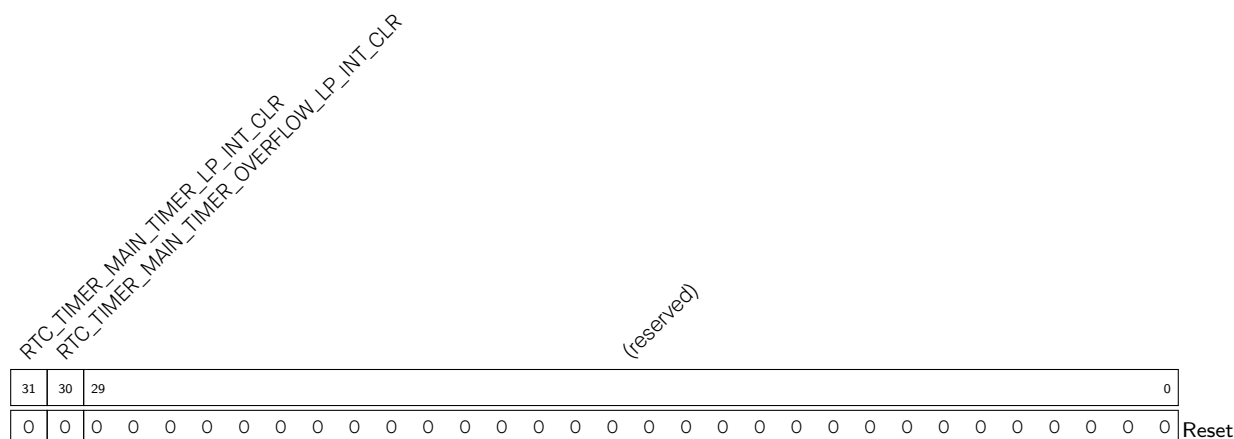
Register 18.16. RTC_TIMER_LP_INT_ENA_REG (0x0040)



RTC_TIMER_MAIN_TIMER_OVERFLOW_LP_INT_ENA Configure to enable the interrupt for the 48-bit counter overflow. (R/W)

RTC_TIMER_MAIN_TIMER_LP_INT_ENA Configure to enable the interrupt for the RTC Timer reaching the target time 1. (R/W)

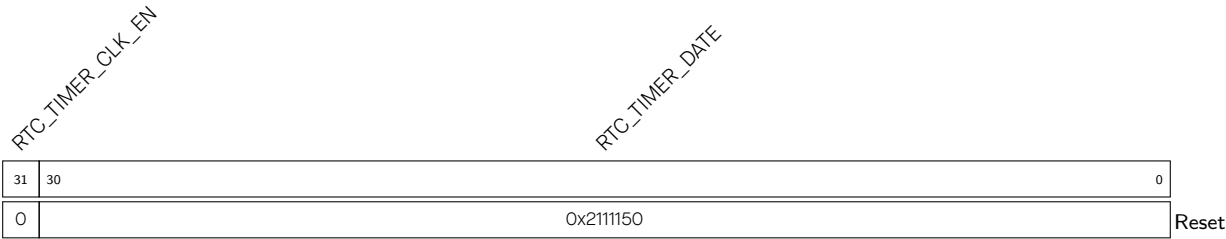
Register 18.17. RTC_TIMER_LP_INT_CLR_REG (0x0044)



RTC_TIMER_MAIN_TIMER_OVERFLOW_LP_INT_CLR Configure to clear the interrupt for 48-bit counter overflow. (WO)

RTC_TIMER_MAIN_TIMER_LP_INT_CLR Configure to clear the interrupt for the RTC Timer reaching the target time 1. (WO)

Register 18.18. RTC_TIMER_DATE_REG (0x03FC)



- RTC_TIMER_DATE** Version control register. (R/W)
- RTC_TIMER_CLK_EN** Configure to enable the register read/write clock. (R/W)